

Smart Touch Panel For Home Automation

Ashwini Kumar Sinha

Traditional switches are outdated now. These switches have mechanical moving parts that get damaged on continuous use.

Many old switchboards are getting replaced by modern touch switches that not only enhance the look of our homes but are easier and safer to use. These modern touch switches have extra functions and, as no moving parts are involved, these last longer than the traditional ones. Despite these advantages, touch switches are

quite expensive.

This smart touch panel project will turn out to be much cheaper than the panels available in the market. The panel can be used to control electrical appliances either through Arduino based touch switchboard or wirelessly through an Android app. Arduino based touch switchboard can also dis-

play room temperature on the touch panel screen (refer Fig. 1).

Now, let's collect the components listed in Table 1 to start the project. TFT touch display shield (6cm) used in this project is shown in Fig. 2.

Setting Arduino library

First of all, install the required libraries in Arduino IDE. For this, go to the Sketch menu and click on Manage libraries option (Fig. 3). Then search the library name and click on the install button. The libraries needed for this project are Adafruit GFX, TouchScreen, and NTC thermistor. Install them, as shown in Fig. 4, Fig. 5, and Fig. 6. Every touchscreen has its own library. Here, we have used MCU-Friend 6cm TFT LCD Library ST7789V.

Coding

After installing the required libraries, let's start Arduino coding.

In the first part of the code, initialise the required libraries, as shown in Fig. 7.

Components Table 1		
Component	Quantity	Description
Arduino Uno	1	For programming
6cm TFT touch display	1	MCUfriend ST7789V
2-channel relay module	1	For switching two devices
Bluetooth HC-05	1	Optional for wireless control
Jumper wires	20	For circuit connections
Switchboard box	1	Enclosure
NTC 103 thermistor	1	Optional for temperature display



Fig. 1: Touch control panel switchboard showing 30°C room temperature



Fig. 2: TFT display for Arduino

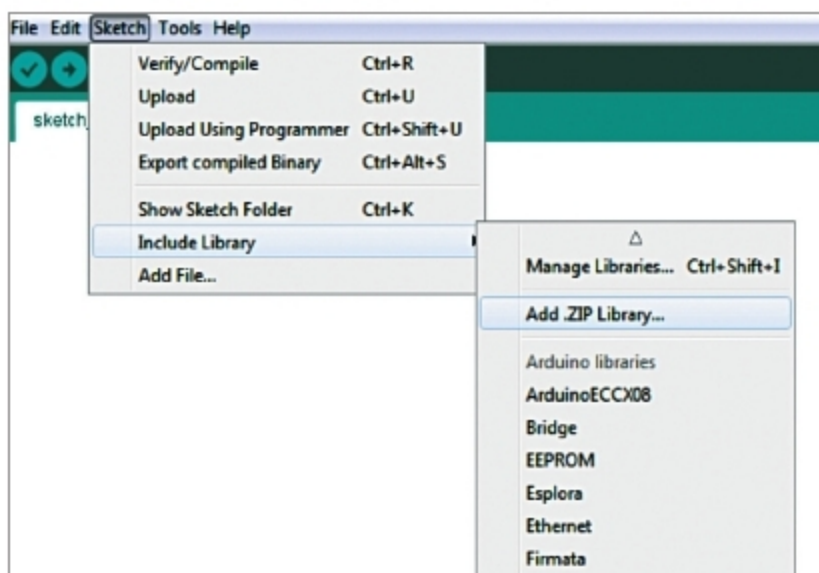


Fig. 3: Opening library manager

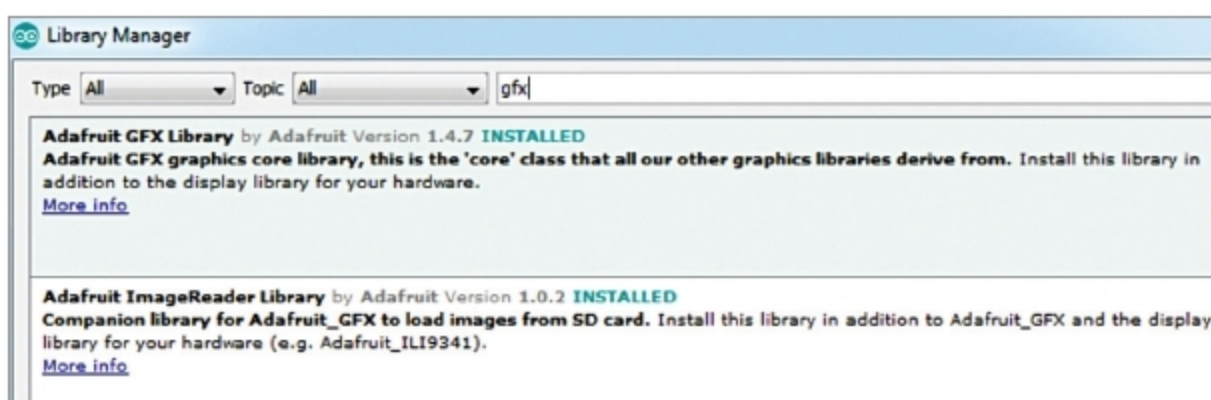


Fig. 4: Installing Adafruit GFX library